

Exploratory Data Analysis Technical Report: Historical US Stock Markets (1970–2018)

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Course: Data Analytics & AI

Scope: In-depth audit, validation, and exploratory analysis of historical equities and trade feeds.

1. Executive Summary

Datasets form the foundation for every analytical pipeline and predictive architecture in applied artificial intelligence. Machine learning models inherit the integrity—or the flaws—of the data that trains them. A model fed uninspected, uncleaned numbers produces distorted forecasts and brittle conclusions. Rigorous data loading, proactive exploration, and disciplined cleaning decisions are mandatory to supply reliable, production-ready information to downstream algorithms.



An analyst must never evaluate data quality based solely on surface-level summary statistics. In financial market analysis, numbers rarely tell their full story at first glance. What appears to be an impossible outlier, or an index-crushing valuation often conceals a distinct market mechanism, such as reverse stock splits or structural non-operating classifications. Conversely, an innocent-looking price record can hide an upstream mathematical overflow. Exploratory Data Analysis (EDA) requires practitioners to stop, read the output, understand what the data communicates, and translate raw figures into accurate representations of real-world financial dynamics.

2. Dataset Overview & Architecture

The project evaluated two primary historical tables spanning four decades of American equity activity:

Metric / Dimension	Securities Master (df_stocks)	Trading Daily Records (df_prices)
Row Count	6,460 rows (6,459 after cleaning)	20,973,889 rows
Column Count	5 attributes	8 attributes
Primary Keys	ticker	Composite: [ticker, date]
Exchanges	NASDAQ (51.22%), NYSE (48.78%)	Multi-Exchange Historical Ticks
Timeframe	Cross-sectional catalog	January 2, 1970 – August 24, 2018
Temporal Span	N/A	17,766 calendar days (~48.6 years)

Converting the raw date strings into native `datetime64[ns]` enabled chronologically accurate time-series indexing, sliding window calculations, and cross-market historical alignments without text-parsing overhead.

3. Data Quality Audit & Forensic Cleaning Decisions

Following the systematic protocol—**Detect -> Investigate -> Document -> Decide**—the audit yielded three primary data quality findings:

- 1. Injected Upstream Header Row:** Row index 74 in `df_stocks` contained the strings `['Symbol', 'NYSE', 'NAME', 'SECTOR', 'INDUSTRY']`, caused by a dirty concatenation during upstream file merging.
 - **Action:** Purged row 74. This prevented string data from polluting ticker foreign keys.
- 2. Missing Sector/Industry Categorizations:** Exactly 1,440 securities (22.29%) lacked sector and industry assignments.
 - **Action:** Retained all 1,440 rows and imputed the category `'Unknown'`.
Dropping these entries would eliminate Exchange-Traded Funds (ETFs), closed-end bond trusts, and Special Purpose Acquisition Companies (SPACs), which do not operate as traditional commercial businesses and carry no standard GICS tags.
- 3. Price Corridor & Logical Violations:** Across ~21 million entries, zero negative or zero-dollar prices existed. However, the sweep identified 48 records where `high < low`, 1,886 records where `open` fell outside the `[low, high]` boundary, and 661 records where `close` fell outside that spread.
 - **Action:** Retained these records (<0.012% of the dataset). These reflect minor data transmission glitches, post-market adjustments, or regional exchange reporting lag. Dropping them would break temporal continuity across 48 years of trading sessions.

4. Numerical Analysis & Distribution Characteristics

Descriptive statistics across the 20,973,889 daily records revealed pronounced positive (right-hand) skewness:

Metric	Open (USD)	High (USD)	Low (USD)	Close (USD)	Volume (Shares)
Mean	76.06	78.04	74.22	76.11	1,227,043.03
Median (50%)	15.45	15.66	15.24	15.45	126,000.00
Std Dev	2,848.40	2,996.75	2,744.82	2,868.94	13,166,864.47
25th Percentile	7.50	7.63	7.36	7.50	22,100.00
75th Percentile	29.72	30.10	29.28	29.72	607,400.00
Maximum	2,034,000.00	2,070,000.00	1,440,000.00	1,779,750.00	4,483,504,400.00

The Mean–Median Divergence

The mean closing price (**76.11 USD**) is nearly five times higher (4.93x) than the median closing price (**15.45 USD**). In standard linear statistics, the mean implies that an average stock trade hovered around 76 USD. In market reality, **75% of all stock sessions across four decades settled below 29.72 USD**. The mean is inflated by high-priced outliers. Relying on median baselines and logarithmic scaling is essential when analyzing historical equities.

5. Forensic Investigation of Extreme Outliers

Investigating the top maximum values revealed the causes behind extreme values in financial datasets:

5.1 Multi-Million-Dollar Raw Closes (Reverse Stock Splits)

- **Observed Anomalies:** `SCON` (Superconductor Technologies) traded at **1,779,750.00 USD** on February 28, 2000, alongside similar values in `CODA` (1,354,500.00 USD), `TVIX` (1,347,500.00 USD), and `TOPS` (1,189,800.00 USD).
- **Diagnosis:** These stocks were not historically expensive blue-chip companies; their daily volumes were tiny (**34 to 400 shares**). These values stem from **repeated reverse stock splits**. When micro-cap companies face exchange delisting for trading under 1.00 USD, they consolidate shares (e.g., 1-for-50 or 1-for-100). When upstream vendors normalize historical share counts retroactively without adjusting raw price histories, pre-split prices mathematically multiply into seven figures.
- **Decision:** Retain the records. They reflect mathematical artifacts from corporate capital structures rather than arbitrary typographical errors.

5.2 The 18.9 Quintillion USD Adjusted Close (Upstream Overflow)

- **Observed Anomaly:** Ticker `AAN` (Aaron's, Inc.) closed normally on February 27, 1987 at **1.30 USD** with 346,900 shares traded. However, its `adj_close` registered at **18,949,618,910,812,438,528.00 USD**—an adjustment multiplier of 14.6 trillion times!
- **Diagnosis:** This is an upstream vendor calculation bug. Adjusted close calculates backwards from current prices, dividing by split ratios and dividend payouts. An unrecorded dividend or a near-zero rounding divisor generated a floating-point overflow. Because the vendor's ingestion pipeline lacked a basic sanity ceiling (e.g., `adj_close <= 500,000`), the corrupted floating-point number exported directly into the CSV feed.
- **Decision:** Do not delete the row. The raw price and volume metrics are accurate. If computing dividend-adjusted returns in future modeling, replace or nullify `adj_close` for `AAN` across Q1 1987 to preserve the valid market data.

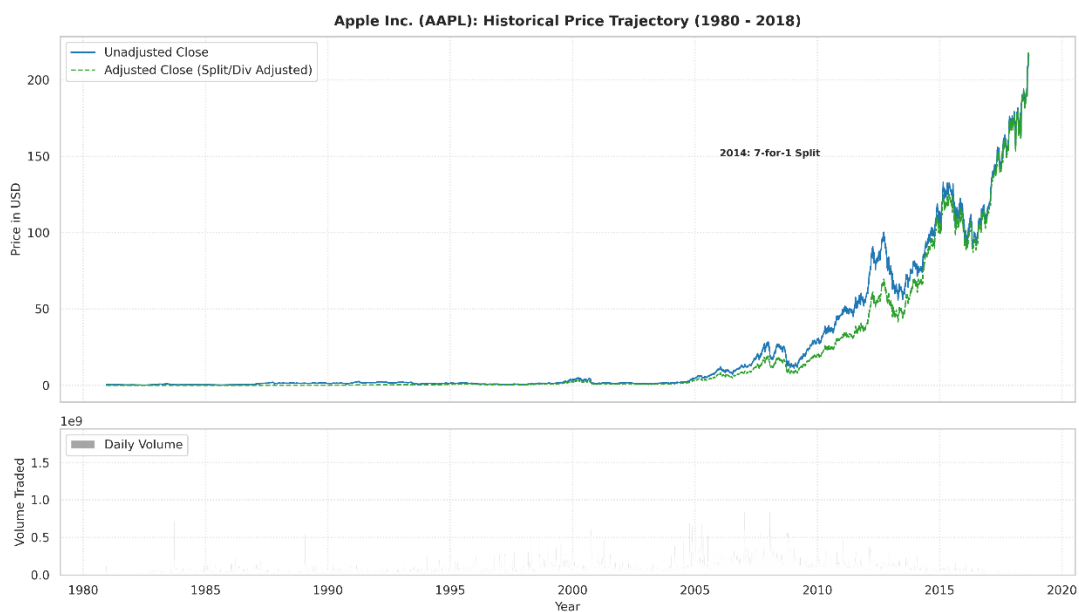
6. Visualization Analysis & Empirical Takeaways

Visualizing the distributions supported the statistical conclusions:

1. **Logarithmic Price Histograms:** Standard histogram bins squeezed 99% of data into a single line against the vertical axis. Applying $\log_{10}$ scaling revealed a clean log-normal distribution centered between 10 USD and 20 USD, with the median (15.45 USD) situated in the central mass and the mean (76.11 USD) pulled along the right tail.
2. **Logarithmic Volume Histograms:** Trading volumes exhibited parallel lognormality. While median volume settled at 126,000 shares, heavy market events created an extended tail reaching single-session volumes of 4.48 billion shares.
3. **Boxplot Outlier Profiles:** The boxplots compressed the interquartile range (IQR) to the bottom of the axis, dominated by a horizontal trail of outlier points stretching beyond 600,000 USD and billions of shares. This confirms that while statistical outliers represent a fraction of a percent of rows, their scale will distort any linear model that lacks robust clipping or scaling.

7. Single Stock Longitudinal Study: Apple Inc. (AAPL)

Evaluating Apple Inc. (AAPL, NASDAQ) isolated macroeconomic patterns into a focused company history:



- **Dataset Timeframe:** December 12, 1980 – August 24, 2018 (9,507 trading sessions).
- **Price Trajectory:** Ranged from an all-time low of **0.20 USD** to an all-time high of **217.58 USD**.
- **Median vs. Mean:** Lifetime median close was **1.65 USD**, while the lifetime mean stood at **24.69 USD**.
- **Analytical Takeaway:** Apple illustrates right-skewness at the individual asset level. For over 24 years (1980 to 2004), the stock traded below 2.00 USD. The exponential run-up driven by the iPhone era (post-2007) and its 7-for-1 stock split in June 2014 created a spread between unadjusted and adjusted prices, lifting the lifetime mean while leaving the median at 1.65 USD.

8. Forward Hypotheses for Predictive Modeling

To extend this exploratory phase into predictive analytics and algorithmic modeling, two target hypotheses are proposed:

- **Hypothesis 1 (Volume Surge as a Forward Volatility Predictor):**
 - *Premise:* When an equity experiences a single-day trading volume expansion exceeding +3.0 standard deviations above its 30-day rolling average, its realized price volatility over the subsequent 20 trading sessions will exceed its historical 90-day baseline by at least 25%.
 - *Application:* Serves as a feature engineering input for automated options-pricing algorithms and risk-management triggers.
- **Hypothesis 2 (Defensive Sector Resilience During Market Corrections):**
 - *Premise:* During systemic macro market contractions (e.g., the 2000–2002 Dot-Com crash and the 2007–2008 Financial Crisis), equities classified under defensive non-cyclical sectors (Health Care, Consumer Non-Durables) will demonstrate lower maximum drawdowns and higher rolling Sharpe ratios than cyclical sectors (Technology, Finance, Capital Goods).
 - *Application:* Direct input for sector-rotation strategies, asset allocation, and algorithmic risk budgeting.